CLOSED POINTS OF FINITE TYPE ARE DENSE IN A DECENT JACOBSON SPACE OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Points lift to closed points. lemma-Jacobson-ft-points-lift-to-closed (carded L26755). The lifting lemma.
- Density after etale cover. Properties of Spaces, lemma etale-image-open (cross-chapter, named). Images of etale morphisms are open.

Source reference. Original definition 03I8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1262–1262. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AHB: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3636–3636. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02J1: focused-seeds/geometry/S9-decent-decent-jacobson-ft-pts/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06EB: focused-seeds/geometry/S9-decent-decent-jacobson-ft-pts/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TB: focused-seeds/geometry/S9-decent-decent-jacobson-ft-pts/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01T8: focused-seeds/geometry/S9-decent-decent-jacobson-ft-pts/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent Jacobson algebraic space over S. Then $X_{ft\text{-}pts} \subset |X|$ is the set of closed points.*

Proof. If $x \in |X|$ is closed, then we can represent x by a closed immersion $\mathrm{Spec}(k) \to X$, see Lemma 0AHB. Hence x is certainly a finite type point.

Conversely, let $x \in |X|$ be a finite type point. We know that x can be represented by a quasi-compact monomorphism $\mathrm{Spec}(k) \to X$ where k is a field (Definition 03I8). On the other hand, by definition, there exists a morphism $\mathrm{Spec}(k') \to X$ which is locally of finite type and represents x (Morphisms, Definition 02J1). We obtain a factorization $\mathrm{Spec}(k') \to \mathrm{Spec}(k) \to X$. Let $U \to X$ be any étale morphism with U affine and consider the morphisms

$$\mathrm{Spec}(k') \times_X U \to \mathrm{Spec}(k) \times_X U \to U$$

The quasi-compact scheme $\mathrm{Spec}(k) \times_X U$ is étale over $\mathrm{Spec}(k)$ hence is a finite disjoint union of spectra of fields (Remark 03II). Moreover, the first morphism is surjective and locally of finite type (Morphisms, Lemma 01T8) hence surjective on finite type points (Morphisms, Lemma 06EB) and the composition (which is locally of finite type) sends finite type points to closed points as U is Jacobson (Morphisms, Lemma 01TB). Thus the image of $\mathrm{Spec}(k) \times_X U \to U$ is a finite set of closed points hence closed. Since this is true for every affine U and étale morphism $U \to X$, we conclude that $x \in |X|$ is closed. □